



9359 - An Extraordinary Strongly-Lensed Ly-C Leaker at $z=4.059$

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Camila Silva (CoI)	University of Chicago
Mr. Daniel Babnigg (CoI)	University of Chicago
Mr. Zewei Wu (CoI)	University of Chicago
Mr. Joseph Yeung (CoI)	University of Chicago

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	1		NIRSpec IFU Spectroscopy	(1) 1402-2809A
	3		NIRSpec IFU Spectroscopy	(2) 1402-2809B
	2		NIRCam Imaging	(1) 1402-2809A

ABSTRACT

Coordinated JWST Observations for HST Cycle 33 proposal 6704:

We propose HST imaging observations to measure the ionizing photons leaking from a remarkable $z=4.059$ strongly lensed galaxy. This galaxy, the highest redshift system yet identified to be spewing ionizing photons into the broader cosmos, and only the third strongly-lensed instance of a Ly-C leaker ever found, offers an opportunity to isolate and study the region(s) within the galaxy that are producing ionizing photons that are escaping. In addition to the direct measurement of the leaking ionizing photons we will also gather multi-band imaging, with both HST and JWST, to measure the continuum spectral energy distributions of all structures within the galaxy. We also propose for initial exploratory JWST spectroscopy to measure restframe emission lines in the nebular gas. All of this imaging and spectroscopy will inform an analysis of the physical conditions within and properties of the leaking components, which be correlated with the leaking Ly-C photons known to be emerging from this galaxy.

OBSERVING DESCRIPTION

The observations are of a single target - a lensed Lyman continuum leaker at $z=4.06$. This is a group-scale lens, forming a lensed "arc" that can be covered in two slightly overlapping NIRSpec-IFU pointings. We are seeking 4-band NirCam imaging, and first-look IFU spectroscopy, sampling the rest frame optical and hence numerous ISM diagnostic lines, which will be analysed in conjunction with the HST-imaging of the leaking UV ionizing photons and the rest-UV continuum.

The total request is 7.8 hours charged time, of which 4.2 hours is science time.

Proposal 9359 - Targets - An Extraordinary Strongly-Lensed Ly-C Leaker at $z=4.059$

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	1402-2809A	RA: 14 02 23.5902 (210.5982925d) Dec: -28 09 35.87 (-28.15996d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Einstein rings, Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies, Lyman-break galaxies]				
Fixed Targets	(2)	1402-2809B	RA: 14 02 23.5759 (210.5982329d) Dec: -28 09 38.56 (-28.16071d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Einstein rings, Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies, Lyman-break galaxies]				
	Extended=YES				

Proposal 9359 - Observation 1 - An Extraordinary Strongly-Lensed Ly-C Leaker at z=4.059

Observation	Proposal 9359, Observation 1											Fri Aug 01 21:00:36 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	1402-2809A	RA: 14 02 23.5902 (210.5982925d) Dec: -28 09 35.87 (-28.15996d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Einstein rings, Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies, Lyman-break galaxies]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235H/F170LP	NRSIRS2	5	1	false	true	NONE	9	9	3413.8	
	2	G395H/F290LP	NRSIRS2	5	1	false	true	NONE	9	9	3413.8	
Special Requirements	Aperture PA Range 70 to 80 Degrees (V3 291.02835083 to 301.02835083) Aperture PA Range 242 to 252 Degrees (V3 103.02835083 to 113.02835083)											

Proposal 9359 - Observation 3 - An Extraordinary Strongly-Lensed Ly-C Leaker at z=4.059

Observation	Proposal 9359, Observation 3 Fri Aug 01 21:00:36 GMT 2025																																															
	Diagnostic Status: Warning Observing Template: NIRSspec IFU Spectroscopy																																															
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.																																															
Fixed Targets	<table border="1"> <thead> <tr> <th>#</th><th>Name</th><th>Target Coordinates</th><th>Targ. Coord. Corrections</th><th>Miscellaneous</th></tr> </thead> <tbody> <tr> <td>(2)</td><td>1402-2809B</td><td> RA: 14 02 23.5759 (210.5982329d) Dec: -28 09 38.56 (-28.16071d) Equinox: J2000 </td><td></td><td></td></tr> </tbody> </table>												#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous	(2)	1402-2809B	RA: 14 02 23.5759 (210.5982329d) Dec: -28 09 38.56 (-28.16071d) Equinox: J2000																												
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Proposal 9359 - Observation 2 - An Extraordinary Strongly-Lensed Ly-C Leaker at z=4.059

Observation	Proposal 9359, Observation 2										Fri Aug 01 21:00:36 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	1402-2809A	RA: 14 02 23.5902 (210.5982925d) Dec: -28 09 35.87 (-28.15996d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Einstein rings, Emission line galaxies, High-redshift galaxies, Lyman-alpha galaxies, Lyman-break galaxies]										
Template	Module			Subarray				Target Placement			
	ALL			FULL				Module Gap			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		6		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F150W	F444W	BRIGHT2	6	1	6	6	773.047	249515.1	
	2	F210M	F300M	BRIGHT2	8	1	6	6	1030.73	249515.2	
Special Requirements	Aperture PA Range 102.95377389 to 112.95377389 Degrees (V3 103.02835083 to 113.02835083) Aperture PA Range 290.95377389 to 300.95377389 Degrees (V3 291.02835083 to 301.02835083) Offset 55.0 arcsec, 35.0 arcsec										